

1. Dataset

The dataset used is **US-Accidents** (Moosavi et al.), available on Kaggle, covering traffic accidents across 49 US states from February 2016 to March 2023. The analysis focuses exclusively on **New York State (NY)**.

Parameter	Original	After preprocessing
Observations	7,730,000	8,448
Variables	43	15
State	49 US states	New York (NY)
City levels	477	125
Weather_Condition levels	47	11

1.1 Variables

Variable	Type	Description
Severity	Continuous	Accident severity score (1–4)
Distance	Continuous	Road extent affected (miles)
City	Categorical	City of accident (125 levels after grouping)
Temperature	Continuous	Temperature in °F
Pressure	Continuous	Atmospheric pressure (inches)
Visibility *	Continuous	Target variable — visibility in miles
Wind_speed	Continuous	Wind speed (mph)
Weather_Condition	Categorical	Weather at time of accident (11 levels after grouping)
Crossing	Binary	Near a pedestrian crossing (True/False)
Junction	Binary	Near an intersection (True/False)
Stop	Binary	Near a stop sign (True/False)
Traffic_signal	Binary	Near a traffic light (True/False)
Sunrise_Sunset	Categorical	Day / Night — also used as binary target
Month	Categorical	Month of accident (1–12)
Day	Categorical	Day of month (1–31)

* Primary target for the linear model.

1.2 Categorical variable preprocessing

City: cities with fewer than 20 recorded accidents were grouped into "Other" and subsequently removed. Levels reduced from **477 to 125**.

Weather_Condition: optimal grouping via the *factorMerger* package (GIC criterion) reduced levels from **47 to 11**.

2. Missing Value Analysis

Missing values were identified using the **VIM** package (*aggr* function). Single imputation was applied: **mean imputation** for continuous variables and **mode imputation** for categorical ones. Correct imputation was verified graphically with a second *aggr* plot.

Variable	Type	Imputation method
wind_speed	Continuous	Mean — Hmisc::impute
temperature	Continuous	Mean — Hmisc::impute
pressure	Continuous	Mean — Hmisc::impute
visibility	Continuous	Mean — Hmisc::impute
Weather_Condition	Categorical	Mode — Hmisc::impute
Sunrise_Sunset	Categorical	Mode — Hmisc::impute

3. Linear Model — Target: Visibility

The response variable is **visibility** (miles), chosen for its relevance to accident risk analysis: understanding how environmental and infrastructure factors affect visibility can inform road safety policy and signage improvements.

1. Model 1 — Full model

All 15 variables included. Non-significant variables: Severity, Crossing, Stop, Traffic_signal, Temperature, Sunrise_Sunset. $R^2 = 0.757$ | AIC = 33,159.

2. Multicollinearity

Quantitative variables: VIF < 5 and TOL > 0.04 — no collinearity detected. Qualitative variables: normalized chi-square < 0.80 for all pairs — no collinearity.

3. Interactions — Model 2

The interaction temperature × Sunrise_Sunset is significant. AIC improves from 33,159 to 33,153 | R^2 from 0.7573 to 0.7575.

4. Y transformation — Box-Cox

Suggested lambda = 1.67. After transformation, AIC worsens from 33,153 to 64,090 and R^2 decreases. Decision: Y not transformed.

5. X transformations — GAM (Model 4)

Loess and Spline fits suggest a cubic relationship for distance and pressure. Cubic terms added. RESET test confirms relevance: p-value < 2.2e-16. $R^2 \rightarrow 0.7598$.

6. Model selection — stepAIC (Model 5)

Bidirectional stepAIC yields a parsimonious model. All retained variables are significant. AIC $\rightarrow 33,079$ | $R^2 = 0.7597$.

7. Outlier removal — DFFITS (Model 6)

Criterion: $|DFFITS| > 2\sqrt{p/n}$. Influential observations removed. R^2 improves substantially: 0.759 $\rightarrow$ 0.8506 | AIC $\rightarrow 26,673$.

8. Heteroscedasticity

Detected via ncvTest and Breusch-Pagan test (p-value = 2.22e-16). Corrected using White's robust standard errors (sandwich package, vcovHC).

9. Bootstrap — R = 1,999 replications

Bootstrap confirms the robustness of all estimated parameters. The final model is stable.

3.1 Model performance summary

Model	R ²	AIC	Notes
Model 1 — full	0.757	33,159	Baseline
Model 2 — + interaction	0.757	33,153	temperature × Sunrise_Sunset
Model 4 — + cubic transforms	0.760	33,083	distance ³ , pressure ³
Model 5 — stepAIC	0.760	33,079	Parsimonious
Model 6 — outlier removal	0.851	26,673	Final model

3.2 Final model specification

The final model (Model 6 with White's robust standard errors):

```

modello_final <- lm(
  visibility ~ distance + I(distance^2) + I(distance^3)
  + City
  + pressure + I(pressure^2) + I(pressure^3)
  + wind_speed + Weather_group + Severity
  + month + day
  + temperature * Sunrise_Sunset,
  data = b_filtered
)

# Robust inference
coeftest(modello_final, vcov = vcovHC(modello_final))

```

R² = 0.8506 | R²_{adj} = 0.8357 | AIC = 26,673.73

4. Logistic Model — Target: Sunrise/Sunset (Day vs Night)

A logistic regression model was built to classify whether an accident occurred **during the day or at night** (variable *Sunrise_Sunset*).

4.1 Modelling pipeline

1. Full logistic model

GLM with family = binomial on all available predictors. Non-significant variables: Junction, Traffic_signal, visibility.

2. Model selection — stepAIC

stepAIC selects model_log2 (AIC = 10,127.52) with predictors: Severity, distance, City, Crossing, Junction, Stop, month, day, wind_speed, temperature, pressure, visibility, Weather_group.

3. ANOVA comparison (LRT)

LRT between logistic1 and model_log2: p-value = 0.5994 > 0.05. Deviance change = -0.326. The simpler model performs equivalently to the full one.

4.2 Odds ratio interpretation

The odds ratio (OR) measures the association between each predictor and the probability that an accident occurred during the day:

Predictor / Group	OR	Interpretation
Amsterdam, Huntington Station, Fresh Meadows	> 1	Strongly increases probability of Day
Weather group 6	> 1	Increases probability of Day
Hudson, Kirkville, Jordan	< 1	Associated with Night accidents
Reduced visibility	< 1	Associated with Night accidents
Wind speed, Temperature	≈ 1	No significant effect

For continuous variables: each one-unit increase changes the odds by a factor of OR. For categorical variables: the odds of Day are OR times higher/lower than the reference category.

5. R Packages

Package	Purpose
dplyr	Data manipulation and filtering
VIM	Missing value visualization (aggr)
Hmisc	Missing value imputation
factorMerger	Optimal grouping of categorical levels
mctest	VIF / TOL for multicollinearity
psych	Pair panels for exploratory analysis
gam	Generalised additive models (Loess, Splines)
lmtest	RESET test and Breusch-Pagan test
MASS	stepAIC and Box-Cox transformation
car	ncvTest and bootstrap (Boot)
sandwich	White's robust standard errors (vcovHC)

6. Replication Instructions

1. Download the US-Accidents dataset from Kaggle (link in Section 1).
2. Filter for State == 'NY' to obtain the New York subset.
3. Set the working directory in *DM_code.R* using `setwd()`.
4. Install all packages listed in Section 5.
5. Run *DM_code.R* sequentially — each block is commented to guide the analysis.

Data Mining course — A.A. 2024/2025 | Dataset: US-Accidents, Sobhan Moosavi et al., Kaggle